Impact energy and TNT equivalents

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Convert an energy estimate without claiming a complete damage prediction.

SUCCESS CRITERIA - I can...

1. Convert speed units before squaring.
2. Calculate kinetic energy from stated model values.
3. Distinguish an energy equivalent from an explosion or damage forecast.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equivalent” and “kinetic energy” to explain your model or result. Sentence starters are optional.

equivalent: equal in the specified quantity, not necessarily in other effects

kinetic energy: energy associated with motion

impact: a collision or strike between objects

01 Bring back an earlier idea

A constant 30 N forward force acts at steady 5 m/s for 7 s. **Find** mechanical power and work.

Word help: power = rate of energy transfer

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$
TNT equivalent (tons) = $K/(4.184\times10^9\ J)$.
1 ton TNT equivalent = 4.184×10^9 J. Equal energy does not imply identical effects.

Worked reasoning: A fictional event transfers 8.4×10^6 J. Using 4.2×10^6 J per kg TNT as an energy comparison gives 2 kg TNT equivalent. This compares energy, not identical effects.

Your turn: Draw a labeled energy comparison and name one effect this number alone cannot predict.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

A report gives impact energy in J and another in TNT-equivalent tons. Are they necessarily describing different physical quantities?

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

04 Solve and justify

Hypothetical mass 10^6 kg at 10 km/s.
Calculate K and TNT equivalent tons.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 5 kg object descends 7 m near Earth.
Use g=10.
Find ΔU and the work done by gravity.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

At fixed speed, mass doubles while assumed energy coupling fraction halves.
Compare coupled energy.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2000 kg hypothetical object moves at 1000 m/s.
Find kinetic energy and kg TNT equivalent using 4.2×10^{6} J/kg.
Explain why neither alone predicts a crater size.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3